

A 5^r -Scaled Family of the Reduced $5k - 1(3)/4$ Collatz-Type Map with Two Observed Attractors up to 10^{11}

Exact Algebraic Conjugacy, Two-Attractor Transport, Orbit-Statistic Invariance, and Transfer of the 10^{11} Finite Verification

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Abstract

We develop the full 5^r -scaled family associated with the residue-dependent reduced $5k - 1(3)/4$ Collatz-type map of the author's base paper. The base map acts on positive odd integers by

$$T_0(y) = \frac{5y - q(y)}{2^{\nu_2(5y - q(y))}}, \quad q(y) = \begin{cases} 1, & y \equiv 1 \pmod{4}, \\ 3, & y \equiv 3 \pmod{4}. \end{cases}$$

For each integer $r \geq 0$ we restrict the state space to $S_r = 5^r \mathbb{N}_{\text{odd}}$ and define

$$M_r(x) = 5x - 5^r q(x), \quad T_r(x) = \frac{M_r(x)}{2^{\nu_2(M_r(x))}}.$$

Because $5^r \equiv 1 \pmod{4}$ and 5^r is odd, multiplication by 5^r preserves both the branch residue and the complete two-adic valuation. We prove the exact conjugacy

$$T_r(5^r y) = 5^r T_0(y), \quad T_r^j(5^r y) = 5^r T_0^j(y).$$

Thus the two displayed base fixed points 1 and 3 become 5^r and $3 \cdot 5^r$, and every periodic orbit in S_r is exactly a 5^r -multiple of a base periodic orbit. Capture times, first-descent stopping times, branch sequences, valuation sequences, basin membership on corresponding finite sets, and normalized orbit profiles are invariant; absolute state and affine peaks scale by 5^r .

The base computation exhaustively verified all 50,000,000,000 positive odd starts below 10^{11} : every tested orbit entered 1 or 3, with zero additional terminal cycles and zero unsigned-128-bit overflow events. We do not present these inherited data as a new scan. Instead, exact conjugacy transfers the complete finite verification to

$$F_r = \{5^r y : 1 \leq y < 10^{11}, y \text{ odd}\} \subset S_r.$$

Numerical examples for several values of r are included solely as direct illustrations of the proved scaling identities. Global two-attractor convergence remains open and is exactly equivalent to the unresolved base conjecture.

Keywords: Collatz-type map; $5x - q$; 5^r scaling; algebraic conjugacy; two-adic valuation; two observed attractors; finite verification; computational number theory.

1 Introduction

The classical $3x + 1$ problem belongs to a broad class of arithmetic dynamical systems in which an affine operation is followed by division by powers of two [1, 2, 3]. The present work concerns a residue-dependent subtractive $5x - q$ system.

The base system is the reduced $5k - 1(3)/4$ domain studied in the author's preceding paper [4]. Its correction is 1 on the odd residue class 1 (mod 4) and 3 on the odd residue class 3 (mod 4). The choice forces at least two factors of two in every affine numerator. The archived base computation exhaustively resolved every positive odd start below 10^{11} and observed exactly the two fixed-point attractors 1 and 3 on that finite range.

Here the question is structural rather than computational: if both the state space and the subtractive correction are multiplied by 5^r , does a genuinely new dynamical system arise? We prove that on the scaled domain $S_r = 5^r \mathbb{N}_{\text{odd}}$ the answer is no. The parameter r gives an exact algebraic scaling symmetry. The present paper is restricted throughout to the subtractive reduced $5k - 1(3)/4$ family and its exact 5^r scaling symmetry.

2 The base reduced $5k - 1(3)/4$ map

Let $S = \mathbb{N}_{\text{odd}} = \{1, 3, 5, \dots\}$. Define

$$q(y) = \begin{cases} 1, & y \equiv 1 \pmod{4}, \\ 3, & y \equiv 3 \pmod{4}, \end{cases} \quad M_0(y) = 5y - q(y), \quad T_0(y) = \frac{M_0(y)}{2^{\nu_2(M_0(y))}}.$$

If $y = 4m + 1$, then $5y - 1 = 4(5m + 1)$; if $y = 4m + 3$, then $5y - 3 = 4(5m + 3)$. Hence $\nu_2(M_0(y)) \geq 2$ and $T_0(y)$ is positive and odd.

The two displayed fixed points are

$$1 \rightarrow 1, \quad 3 \rightarrow 3,$$

since $5 - 1 = 4$ and $15 - 3 = 12 = 4 \cdot 3$. Some complete base trajectories are

$$\begin{aligned} 5 &\rightarrow 3, & 9 &\rightarrow 11 \rightarrow 13 \rightarrow 1, \\ 27 &\rightarrow 33 \rightarrow 41 \rightarrow 51 \rightarrow 63 \rightarrow 39 \rightarrow 3, \end{aligned}$$

and

$$31 \rightarrow 19 \rightarrow 23 \rightarrow 7 \rightarrow 1.$$

These examples will later be scaled to illustrate the conjugacy.

3 The full 5^r -scaled subtractive family

Fix $r \geq 0$.

3.1 Scaled domain

Define

$$S_r = 5^r S = \{5^r y : y \in S\}.$$

Every $x \in S_r$ has a unique form $x = 5^r y$ with y odd. Since $5 \equiv 1 \pmod{4}$,

$$5^r \equiv 1 \pmod{4},$$

and consequently $5^r y \equiv y \pmod{4}$ and $q(5^r y) = q(y)$.

3.2 Scaled affine and reduced maps

For $x \in S_r$ define

$$M_r(x) = 5x - 5^r q(x) = \begin{cases} 5x - 5^r, & x \equiv 1 \pmod{4}, \\ 5x - 3 \cdot 5^r, & x \equiv 3 \pmod{4}, \end{cases}$$

and

$$T_r(x) = \frac{M_r(x)}{2^{\nu_2(M_r(x))}}.$$

The forced first-compression forms are therefore

$$\frac{5x - 5^r}{4} \quad (x \equiv 1 \pmod{4}), \quad \frac{5x - 3 \cdot 5^r}{4} \quad (x \equiv 3 \pmod{4}),$$

followed by removal of every additional factor of two.

Remark 1 (Why the scaled domain is essential). The exact conjugacy is a theorem on $S_r = 5^r \mathbb{N}_{\text{odd}}$. It is not a statement about all positive odd integers under the displayed affine formulas. The factorization used below requires $x = 5^r y$.

4 Key scaling identities

Lemma 1 (Affine scaling). *For every $r \geq 0$ and $y \in S$,*

$$M_r(5^r y) = 5^r M_0(y).$$

Proof. Because $q(5^r y) = q(y)$,

$$M_r(5^r y) = 5(5^r y) - 5^r q(y) = 5^r(5y - q(y)) = 5^r M_0(y).$$

□

Lemma 2 (Valuation invariance). *For every nonzero integer n and $r \geq 0$,*

$$\nu_2(5^r n) = \nu_2(n).$$

Proof. The factor 5^r is odd and contributes no factor of two. □

Thus corresponding states have exactly the same removed power of two at every matched step.

5 Exact algebraic conjugacy

Define $\Phi_r : S \rightarrow S_r$ by $\Phi_r(y) = 5^r y$.

Theorem 1 (One-step conjugacy). *For every $r \geq 0$ and $y \in S$,*

$$T_r(5^r y) = 5^r T_0(y).$$

Equivalently, $T_r \circ \Phi_r = \Phi_r \circ T_0$.

Proof. Using the two lemmas,

$$T_r(5^r y) = \frac{5^r M_0(y)}{2^{\nu_2(M_0(y))}} = 5^r T_0(y).$$

□

Theorem 2 (All iterates). *For every $j \geq 0$,*

$$T_r^j(5^r y) = 5^r T_0^j(y).$$

Proof. Induction on j , using the one-step identity. □

This identity proves orbit-by-orbit equivalence, not merely statistical similarity.

6 The two displayed attractors under scaling

Theorem 3 (Periodic-orbit bijection). *A tuple $C = (c_0, \dots, c_{L-1})$ is a periodic orbit of T_0 if and only if $5^r C = (5^r c_0, \dots, 5^r c_{L-1})$ is a periodic orbit of T_r . Minimal period is preserved.*

Proof. Apply the all-iterate identity in both directions through the bijection Φ_r . □

Corollary 1. *The two displayed base fixed points become*

$$5^r \rightarrow 5^r, \quad 3 \cdot 5^r \rightarrow 3 \cdot 5^r.$$

No additional periodic orbit can occur inside S_r unless a corresponding periodic orbit exists in the base system.

Directly,

$$5(5^r) - 5^r = 4 \cdot 5^r,$$

and

$$5(3 \cdot 5^r) - 3 \cdot 5^r = 12 \cdot 5^r = 4(3 \cdot 5^r).$$

Thus both scaled fixed points have valuation 2.

7 Numerical illustrations of exact conjugacy

These examples are algebraic illustrations generated from the proved identity; they are not new large-scale computational evidence.

For $r = 1$, the fixed points are 5 and 15. Scaling the base trajectories by 5 gives

$$25 \rightarrow 15,$$

$$45 \rightarrow 55 \rightarrow 65 \rightarrow 5,$$

$$135 \rightarrow 165 \rightarrow 205 \rightarrow 255 \rightarrow 315 \rightarrow 195 \rightarrow 15,$$

and

$$155 \rightarrow 95 \rightarrow 115 \rightarrow 35 \rightarrow 5.$$

For example, at $x = 45 \equiv 1 \pmod{4}$,

$$M_1(45) = 5(45) - 5 = 220 = 4 \cdot 55,$$

so $T_1(45) = 55 = 5T_0(9)$. At $x = 135 \equiv 3 \pmod{4}$,

$$M_1(135) = 675 - 15 = 660 = 4 \cdot 165.$$

For $r = 2$, the fixed points are 25 and 75, and

$$125 \rightarrow 75,$$

$$225 \rightarrow 275 \rightarrow 325 \rightarrow 25,$$

$$675 \rightarrow 825 \rightarrow 1025 \rightarrow 1275 \rightarrow 1575 \rightarrow 975 \rightarrow 75.$$

For $r = 3$, the fixed points are 125 and 375, while the base orbit $31 \rightarrow 19 \rightarrow 23 \rightarrow 7 \rightarrow 1$ becomes

$$3875 \rightarrow 2375 \rightarrow 2875 \rightarrow 875 \rightarrow 125.$$

Each displayed arrow may be checked either directly from T_r or by division by 5^r , application of T_0 , and rescaling.

8 Periodic-orbit identity under scaling

For a base orbit write

$$2^{a_i} y_{i+1} = 5y_i - q_i, \quad a_i = \nu_2(5y_i - q_i) \geq 2,$$

where $q_i \in \{1, 3\}$. For the scaled orbit $x_i = 5^r y_i$,

$$2^{a_i} x_{i+1} = 5x_i - 5^r q_i.$$

Let $A = \sum_{i=0}^{L-1} a_i$ and $S_j = \sum_{i=0}^{j-1} a_i$, $S_0 = 0$. Successive substitution around a length- L cycle gives

$$(5^L - 2^A)x_0 = 5^r \sum_{j=0}^{L-1} 5^{L-1-j} q_j 2^{S_j}.$$

Hence

$$x_0 = 5^r \frac{\sum_{j=0}^{L-1} 5^{L-1-j} q_j 2^{S_j}}{5^L - 2^A} = 5^r y_0.$$

Because the right-hand numerator is positive, every positive periodic orbit must satisfy

$$5^L > 2^A, \quad \frac{A}{L} < \log_2 5.$$

Thus scaling creates no new Diophantine integrality condition.

9 Orbit statistics transported by conjugacy

Let $y_j = T_0^j(y)$ and $x_j = T_r^j(5^r y)$. Then $x_j = 5^r y_j$ for all j .

9.1 Capture and first-descent times

If $\tau_0(y)$ is the first entrance time into $\{1, 3\}$, then

$$\tau_r(5^r y) = \tau_0(y).$$

If

$$\sigma_0(y) = \min\{j \geq 1 : T_0^j(y) < y\},$$

then multiplication by the positive constant 5^r preserves inequalities and

$$\sigma_r(5^r y) = \sigma_0(y).$$

The first-lower value itself is multiplied by 5^r .

9.2 Peaks and normalized profiles

For every matched finite segment,

$$\max_j x_j = 5^r \max_j y_j.$$

Raw affine numerators scale in the same way. However,

$$\frac{x_j}{x_0} = \frac{y_j}{y_0},$$

so normalized orbit profiles, peak/start ratios, and peak iteration indices are invariant.

9.3 Basin membership

If $B_0(1)$ and $B_0(3)$ are the base basins and $B_r(5^r)$ and $B_r(3 \cdot 5^r)$ are their counterparts within S_r , then

$$5^r y \in B_r(5^r) \iff y \in B_0(1),$$

$$5^r y \in B_r(3 \cdot 5^r) \iff y \in B_0(3).$$

Thus counts and proportions are exactly preserved on corresponding finite sets.

Quantity	Behavior under $y \mapsto 5^r y$
Branch sequence	invariant
Two-adic valuation sequence	invariant
Capture time	invariant
First-descent stopping time	invariant
Basin membership	transported bijectively
Peak iteration index	invariant
Normalized state x_j/x_0	invariant
Reduced state values	multiplied by 5^r
First-lower value	multiplied by 5^r
Raw affine numerator	multiplied by 5^r
Absolute reduced peak	multiplied by 5^r

10 Two-adic law and drift under scaling

The base residue model has

$$\Pr(a = k) = 2^{-(k-1)}, \quad k \geq 2, \quad \mathbb{E}[a] = 3.$$

For large states the logarithmic increment is approximately

$$\log 5 - a \log 2,$$

so

$$\mathbb{E}[\Delta \log] \approx \log 5 - 3 \log 2 = \log(5/8) < 0.$$

At matched states the normalized one-step ratio is exactly invariant:

$$\frac{T_r(5^r y)}{5^r y} = \frac{T_0(y)}{y}.$$

Therefore the scaled family inherits the same valuation model, critical mean $\log_2 5$, and negative-logarithmic-drift heuristic. This remains probabilistic evidence, not a global convergence proof.

11 Exact inverse branches and scaled inverse forests

Suppose $T_0(y) = z$ and the removed power is 2^a , $a \geq 2$. The base inverse candidates are

$$y = \frac{2^a z + 1}{5} \quad (q = 1), \quad y = \frac{2^a z + 3}{5} \quad (q = 3),$$

subject to integrality and the appropriate residue branch. Their congruence conditions are

$$2^a z \equiv 4 \pmod{5}, \quad 2^a z \equiv 2 \pmod{5}.$$

For $w = 5^r z$ the scaled inverse candidates are

$$x = \frac{2^a w + 5^r}{5} = 5^r \frac{2^a z + 1}{5},$$

and

$$x = \frac{2^a w + 3 \cdot 5^r}{5} = 5^r \frac{2^a z + 3}{5}.$$

Hence every inverse edge and both inverse forests are exact 5^r -scaled copies of the base structures.

12 Finite verification inherited from the base paper

The base computation [4] exhaustively tested every positive odd start below 10^{11} , a total of 50,000,000,000 starts. Every tested orbit reached 1 or 3; the other-cycle count and unsigned-128-bit overflow count were both zero. The exact finite basin counts were

$$32,318,909,987 \quad \text{for } 1, \quad 17,681,090,013 \quad \text{for } 3,$$

corresponding to 64.637819974% and 35.362180026%.

For every fixed r , define

$$F_r = \{5^r y : 1 \leq y < 10^{11}, \ y \text{ odd}\}.$$

It contains exactly 50,000,000,000 starts. These are inherited trajectories, not newly recomputed measurements of the scaled family.

Theorem 4 (Finite transfer theorem). *For every $r \geq 0$, every $x = 5^r y \in F_r$ enters either 5^r or $3 \cdot 5^r$, under exactly the same finite computational conditions as the base verification. The two basin counts are unchanged on F_r .*

Proof. For each corresponding y , the base computation supplies the finite trajectory statement. The all-iterate conjugacy transports that entire trajectory by multiplication by 5^r . $\square$

The base maximum terminal stopping time was 192 at $y = 86,568,413,515$, and the maximum first-lower stopping time was 102 at $y = 25,450,479,209$. These iteration counts remain unchanged on F_r , while the corresponding starts are multiplied by 5^r . The base maximum reduced odd peak was 303,776,985,040,453, from start 62,483,301,665, and the largest raw affine numerator was 1,518,884,925,202,264; their scaled counterparts are multiplied by 5^r . These values are inherited consequences of the base record, not new scans.

13 Global equivalence and the status of the two attractors

Conjecture 1 (Scaled Two-Attractor Conjecture). *For every $r \geq 0$ and every $x \in S_r$, the orbit of x under T_r eventually enters exactly one of the two fixed points 5^r or $3 \cdot 5^r$.*

Theorem 5 (Equivalence of global conjectures). *For any fixed $r \geq 0$, the scaled conjecture on S_r holds if and only if the base Two-Attractor Conjecture holds on S . Hence it holds for every r if and only if it holds for $r = 0$.*

Proof. The bijective conjugacy Φ_r transports every orbit in both directions. $\square$

Thus the phrase “two observed attractors up to 10^{11} ” must not be confused with a proof that no further attractor or divergent orbit exists globally.

14 What is proved and what is not

The exact algebra proves, for every $r \geq 0$: the bijection $S \leftrightarrow S_r$; preservation of the branch residue; affine scaling; full two-adic valuation invariance; exact conjugacy; correspondence of all iterates and periodic orbits; preservation of minimal periods, capture times and first-descent times; scaling of absolute state values and peaks; invariance of normalized orbit profiles; transport of basin membership; and exact scaling of inverse forests.

The archived computation proves a finite statement for the base starts below 10^{11} . Conjugacy then proves the corresponding finite statement on F_r . What is not proved is global convergence of all positive base or scaled orbits, nonexistence of additional cycles outside the verified finite base range, or existence and exact values of limiting natural basin densities.

15 Reproducibility and verification

The accompanying computational pack contains this L^AT_EX source, a small independent Python consistency verifier, a machine-readable JSON statement of the scaling rules and inherited base statistics, a README, and SHA-256 checksums. The verifier checks, on user-selected finite ranges,

$$M_r(5^r y) = 5^r M_0(y), \quad \nu_2(M_r(5^r y)) = \nu_2(M_0(y)), \quad T_r(5^r y) = 5^r T_0(y),$$

and compares several iterates and the two scaled fixed points. The program is a consistency check; the proof is the algebraic factorization together with $5^r \equiv 1 \pmod{4}$ and $\gcd(5^r, 2) = 1$.

16 Conclusion

Scaling both the state space and the subtractive correction by 5^r produces an exact copy of the reduced $5k - 1(3)/4$ dynamics on S_r . The two affine branches are $5x - 5^r$ and $5x - 3 \cdot 5^r$, with forced first division by 4 and subsequent complete two-adic reduction. The central identity

$$T_r(5^r y) = 5^r T_0(y)$$

transports every orbit exactly. The two displayed fixed points are 5^r and $3 \cdot 5^r$; orbit times and valuation sequences are invariant, while absolute state magnitudes scale by 5^r .

The exhaustive base verification of all fifty billion positive odd starts below 10^{11} therefore transfers exactly to the corresponding structured set F_r , without being relabeled as a new scan. The global convergence problem is unchanged: a proof or counterexample for the base system propagates through the entire scaled family. Thus r is an exact scaling symmetry, not a source of independent dynamics.

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